



mental conditions, gestures might be really useful. For example, a scuba diver could do with a robot assistant that understands gestures. He or she could point at objects and instruct the robot to click photographs from specific angles.

Or, let us say a health-awareness video is being shown to a group of illiterate people in a village. The viewers are not tech-savvy, the environment is likely to be noisy and the people might even speak different dialects. If the computer could understand their gestures to carry out instructions like replay or volume control, it might improve the effectiveness of the presentation.

When you are on a snowy peak wearing heavy gloves, it might not be easy to use a touchscreen. Gesture computing could be a life saver then. Baby monitors, computing systems for people with disabilities, support systems for people working under dangerous environments and more could also work with gesture systems.

Let us look at some projects that aim to improve the use of gestures in such specialised applications.

Working with the tongue

When we think of gesture based computing, the first thing that comes to mind is people with sensor-studded gloves waving their hands this way and that. But how do people with paralysis communicate with their computer, when it is not possible for them to move their arms or legs? Such patients can only use head, tongue and eye movements as inputs for gesture-interface systems. Of these, tongue based systems are popular for paralysis patients because tongue is a flexible organ that can be used to perform several gestures. It is also considered to be intuitive and more reliable because the tongue is controlled by the cranial nerve, which is embedded in the skull and is not easily damaged during spinal cord injuries.

However, current tongue based

Do gloves irk you

Well, even Tom Cruise wore gloves in *Minority Report*. But you might not have to. Gesture interface start-up UltraHaptics has developed a ready-to-test technology for users to receive tactile feedback without wearing any special devices or touching any surfaces. The technology helps them feel menu options in air, interact with virtual objects or get feedback for mid-air gestures, by using ultrasound to project sensations through the air and directly onto the users.

The company was founded in 2013 to build on a technology originally developed at University of Bristol, UK. They are now working with Tier-1 customers to evaluate their solution. UltraHaptics was co-founded by Sriram Subramanian, a professor of informatics at University of Sussex, UK, with research interests in exploring new forms of interactive systems.

input systems are very intrusive. These often require magnetic sensors to be built into dentures or artificial teeth, or for contact with the skin using electromyography (EMG) sensors. This can be quite uncomfortable as EMG electrodes can cause skin abrasions in the long run.

To address this problem, researchers at University of Maryland, Baltimore County, USA, recently proposed a novel contactless sensor called Tongue-n-Cheek, which uses micro-radars to capture tongue gestures. The prototype system is in the form of a helmet to be worn by the user. It comprises a sensor array with three micro-radars bolted to thinly-cut acrylic sheets attached to the helmet.

The team has fabricated a custom-designed printed circuit board (PCB) to house a filtering circuit, an amplifier and an analogue-to-digital converter. The micro-radars act as proximity sensors to capture muscle movements when the patient performs tongue gestures. Tongue-n-Cheek system then converts these movements into gestures using a novel signal-processing algorithm. Tests have showed that the system could decipher tongue gestures with 95 per cent accuracy and low latency.

Wish you a safe drive

Not having to fumble with buttons and levers would be a huge help for drivers and, understandably, automakers are exploring the use of gestures to control car functions. This was one of the prominent themes

at this year's Consumer Electronics Show (CES 2016) held at Las Vegas, USA, as well.

Intuitive infotainment. BMW showcased a few gesture controls in its 7-series, achieved through sensors on the cabin roof, monitoring the area near the infotainment panel. The controls, which experts have reviewed favourably, enable you to change the volume through a circular motion of the index finger, answer or dismiss a call by pointing straight or swiping right, and configure one other setting of your choice by swiping down two fingers.

BMW also showcased AirTouch, a futuristic UI implemented in its prototype Vision Car. Building on the existing gesture-recognition system, AirTouch lets both driver and passengers to control the car's features with gestures using sensors that record 3D hand movements in the area ahead of the infotainment panel. The display is also rather large and panoramic, enabling users to point to required menu choices more easily.

AirTouch is intuitive, in the sense that it attempts to anticipate the next command and reduces the steps required by the driver to complete a task, with minimal distractions. For example, as soon as the driver activates the phone with a simple gesture, the system automatically opens the contacts list so that a call can be placed easily.

Volkswagen also demonstrated some gesture features like controlling the infotainment panel, opening and closing doors and more.